

ENGR 213: Applied Ordinary Differential Equations

Lecture 5 Explained · Solutions by Substitutions

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1. Executive Summary & The Transformation Philosophy

In previous lectures, three core analytical techniques were developed for first-order ordinary differential equations:

1. **Separable Equations:** $\frac{dy}{dx} = g(x)h(y) \implies \frac{dy}{h(y)} = g(x)dx$
2. **Linear Equations:** $\frac{dy}{dx} + P(x)y = Q(x) \implies \mu(x) = e^{\int P(x)dx}$
3. **Exact Equations:** $M(x, y)dx + N(x, y)dy = 0$ with $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Many real-world engineering models do not immediately fit these three forms. However, a large family of non-linear equations can be transformed into **separable** or **linear** equations via a strategic change of variables (substitution).

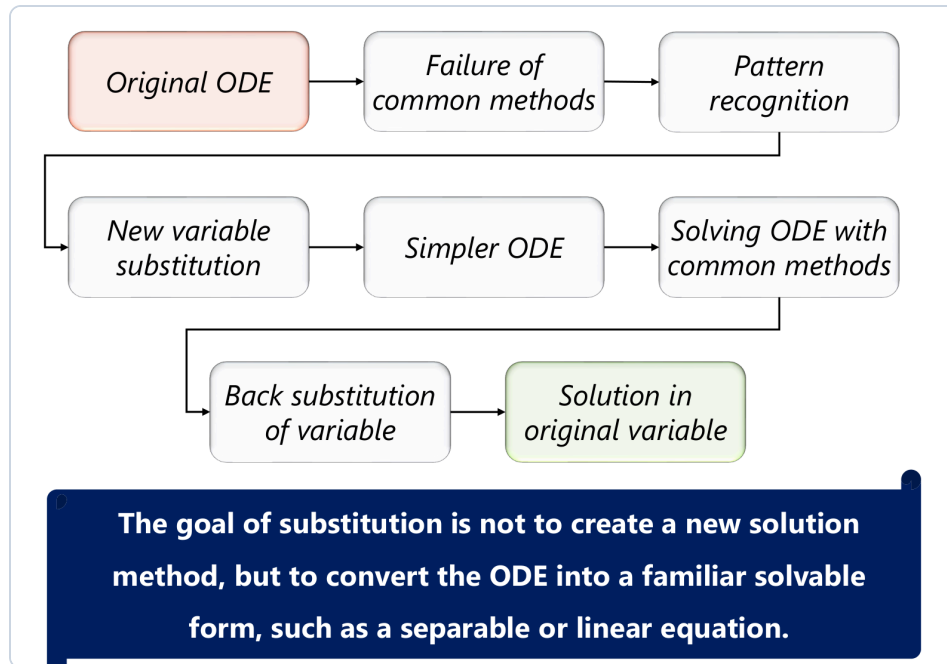


Figure 1: The substitution strategy, Lecture 5, slide 6.

Reading the flowchart: start from the **original ODE**. When the common methods (separable, linear, exact) **fail**, look for a **pattern** that suggests a **new variable** (the substitution). The substitution turns it into a **simpler ODE**, which you solve with a common method; finally **back-substitute** to get the solution in the original variables. The blue box states the purpose: *the goal of substitution is not to create a new solution method, but to convert the ODE into a familiar solvable form, such as a separable or linear equation.*

Pattern in the ODE	Substitution	Becomes
M and N homogeneous of the same degree	$y = ux$ (or $x = vy$)	Separable in u and x
$y' + P(x)y = f(x)y^n, n \neq 0, 1$ (Bernoulli)	$u = y^{1-n}$	Linear in u
$y' = f(Ax + By + C)$	$u = Ax + By + C$	Separable in u and x

[!IMPORTANT] The Core Rule: The goal of substitution is never to create an entirely new calculus method; it is strictly an algebraic bridge that converts an intractable non-linear problem into an elementary separable or linear ODE that we already know how to solve.

2. Pattern Recognition & Diagnostic Framework

Before computing derivatives, an engineer must instantly recognize which substitution applies by inspecting the mathematical structure of the equation.

The Introductory Prototype (Lecture 5 Slide 4)

Consider the differential equation:

$$\frac{dy}{dx} = 1 + \frac{y}{x} + \left(\frac{y}{x}\right)^2$$

- This equation is **not linear** (due to $(y/x)^2$).
- It is **not separable** in x and y as written.
- However, the variable combinations appear exclusively as the ratio $\frac{y}{x}$.
- By substituting $u = \frac{y}{x} \iff y = ux$, direct differentiation gives $\frac{dy}{dx} = u + x \frac{du}{dx}$.
- Substituting this into the ODE collapses it into a clean separable equation:

$$u + x \frac{du}{dx} = 1 + u + u^2 \implies x \frac{du}{dx} = 1 + u^2 \implies \frac{du}{1 + u^2} = \frac{dx}{x}$$

$$\arctan(u) = \ln|x| + C \implies u = \tan(\ln|x| + C) \implies \mathbf{y(x) = x \tan(\ln|x| + C)}$$

Example 1: Classification & Substitution Choice (Lecture 5 Slide 7)

Determine the appropriate substitution type and change of variable for each given ODE:

(a) $\frac{dy}{dx} = (x + y)^2$

- **Pattern:** The right-hand side is a function of the linear combination $Ax + By + C$ with $A = 1, B = 1, C = 0$.
- **Classification:** Reduction to Separation of Variables.
- **Substitution:** Let $u = x + y \implies \frac{du}{dx} = 1 + \frac{dy}{dx} \implies \frac{dy}{dx} = \frac{du}{dx} - 1$.
- **Transformed ODE:** $\frac{du}{dx} - 1 = u^2 \implies \frac{du}{dx} = u^2 + 1$ (Separable).
- **Integration:** $\frac{du}{u^2+1} = dx \implies \arctan(u) = x + C \implies u = \tan(x + C)$.
- **Back-Substitution:** $x + y = \tan(x + C) \implies \mathbf{y(x) = \tan(x + C) - x}$.

(b) $\frac{dy}{dx} = 2 + e^{2x-y+1}$

- **Pattern:** The exponent contains the linear form $Ax + By + C$ with $A = 2, B = -1, C = 1$.
- **Classification:** Reduction to Separation of Variables.
- **Substitution:** Let $u = 2x - y + 1 \implies \frac{du}{dx} = 2 - \frac{dy}{dx} \implies \frac{dy}{dx} = 2 - \frac{du}{dx}$.
- **Transformed ODE:** $2 - \frac{du}{dx} = 2 + e^u \implies \frac{du}{dx} = -e^u$ (Separable).
- **Integration:** $e^{-u} du = -dx \implies -e^{-u} = -x + c_1 \implies e^{-u} = x + c$.
- **Back-Substitution:** $-u = \ln|x + c| \implies 2x - y + 1 = -\ln|x + c| \implies \mathbf{y(x) = 2x + 1 + \ln|x + c|}$.

(c) $\frac{dy}{dx} + 2y = xy^3$

- **Pattern:** Matches the canonical Bernoulli form $\frac{dy}{dx} + P(x)y = f(x)y^n$ with $P(x) = 2, f(x) = x, n = 3$.
- **Classification:** Bernoulli's Differential Equation.
- **Substitution:** Let $u = y^{1-n} = y^{1-3} = y^{-2} = \frac{1}{y^2}$.

- **Transformed ODE:** $\frac{du}{dx} + (1 - n)P(x)u = (1 - n)f(x) \implies \frac{du}{dx} - 4u = -2x$ (Linear in u).

(d) $\frac{dy}{dx} = y(x + \ln y)$

- **Pattern:** Rewrite as $\frac{1}{y} \frac{dy}{dx} - \ln y = x$.
- **Classification:** **Logarithmic Substitution (Special Linear Form).**
- **Substitution:** Let $v = \ln y \implies \frac{dv}{dx} = \frac{1}{y} \frac{dy}{dx}$.
- **Transformed ODE:** $\frac{dv}{dx} - v = x$ (Standard 1st-order linear ODE with $\mu(x) = e^{-x}$).

3. Method 1: Homogeneous Equations

Definition of Homogeneous Functions

A function $f(x, y)$ is said to be **homogeneous of degree α** if scaling both independent inputs by a parameter t satisfies:

$$f(tx, ty) = t^\alpha f(x, y) \quad \text{for all real } t > 0$$

Verification Tests:

- $f(x, y) = x^3 + y^3 \implies f(tx, ty) = (tx)^3 + (ty)^3 = t^3(x^3 + y^3) = t^3 f(x, y) \implies$ **Homogeneous of degree 3.**
- $f(x, y) = x^2 + 5xy \implies f(tx, ty) = t^2 x^2 + 5(tx)(ty) = t^2(x^2 + 5xy) = t^2 f(x, y) \implies$ **Homogeneous of degree 2.**
- $f(x, y) = x^2 + y \implies f(tx, ty) = t^2 x^2 + ty \neq t^\alpha f(x, y) \implies$ **NOT Homogeneous.**

Differential Form Homogeneity Criterion

A first-order differential equation expressed in differential form:

$$M(x, y)dx + N(x, y)dy = 0$$

is **homogeneous** if both coefficient functions $M(x, y)$ and $N(x, y)$ are homogeneous functions of the **exact same degree α** .

Equivalently, writing in derivative form:

$$\frac{dy}{dx} = -\frac{M(x, y)}{N(x, y)} = f\left(\frac{y}{x}\right)$$

The right-hand side can be expressed purely as a function of the single combined variable $\frac{y}{x}$.

The Algebraic Transformations: $y = ux$ vs. $x = vy$

Either substitution will reduce any homogeneous equation to a separable equation:

Criterion	Primary Substitution: $y = ux$	Alternative Substitution: $x = vy$
When to Choose	When $N(x, y)$ is simpler than $M(x, y)$	When $M(x, y)$ is simpler than $N(x, y)$
Differential Relation	$dy = u dx + x du$	$dx = v dy + y dv$
Variable Reduced	Replaces y and dy throughout	Replaces x and dx throughout
Resulting ODE Form	Separable in u and x	Separable in v and y

Example 2: Complete Derivation (Lecture 5 Slide 9)

Problem Statement

Solve the first-order differential equation:

$$(x^2 + y^2)dx + (x^2 - xy)dy = 0$$

Step-by-Step Solution

Step 1: Test for Homogeneity

Identify $M(x, y)$ and $N(x, y)$:

$$M(x, y) = x^2 + y^2 \implies M(tx, ty) = t^2(x^2 + y^2) = t^2 M(x, y) \quad (\alpha = 2)$$

$$N(x, y) = x^2 - xy \implies N(tx, ty) = (tx)^2 - (tx)(ty) = t^2(x^2 - xy) = t^2 N(x, y) \quad (\alpha = 2)$$

Both M and N are homogeneous of degree 2. The equation is **homogeneous**.

Step 2: Choose the Optimal Substitution

Comparing M and N :

- $M(x, y) = x^2 + y^2$ (a sum of two squares)
- $N(x, y) = x^2 - xy = x(x - y)$ (factors easily) Since $N(x, y)$ factors into a single term times $(x - y)$, substituting for dy via $y = ux$ will cancel cleanly. Let:

$$y = ux \implies dy = u dx + x du$$

Step 3: Substitute into the Differential Equation

$$(x^2 + (ux)^2)dx + (x^2 - x(ux))(u dx + x du) = 0$$

Factor out x^2 :

$$x^2(1 + u^2)dx + x^2(1 - u)(u dx + x du) = 0$$

Assuming $x \neq 0$, divide both sides by x^2 :

$$(1 + u^2)dx + (1 - u)(u dx + x du) = 0$$

Expand the second term:

$$(1 + u^2)dx + (u - u^2)dx + x(1 - u)du = 0$$

Combine the dx terms:

$$[(1 + u^2) + (u - u^2)]dx + x(1 - u)du = 0$$

$$(1 + u)dx + x(1 - u)du = 0$$

Step 4: Separate the Variables

Rearrange terms to separate x and u :

$$x(1 - u)du = -(1 + u)dx$$

Divide by $x(1 + u)$ (for $x \neq 0, u \neq -1$):

$$\frac{1 - u}{1 + u}du = -\frac{1}{x}dx \implies \frac{dx}{x} + \frac{u - 1}{u + 1}du = 0$$

Step 5: Perform Polynomial Division on the u -Fraction

$$\frac{u - 1}{u + 1} = \frac{(u + 1) - 2}{u + 1} = 1 - \frac{2}{u + 1}$$

Substitute back into the separated equation:

$$\frac{dx}{x} + \left(1 - \frac{2}{u+1}\right) du = 0$$

Step 6: Integrate Both Sides

$$\int \frac{dx}{x} + \int \left(1 - \frac{2}{u+1}\right) du = C_1$$

$$\ln|x| + u - 2\ln|u+1| = C_1$$

Combine logarithms using log rules:

$$\ln\left|\frac{x}{(u+1)^2}\right| + u = C_1$$

$$\ln\left|\frac{x}{(u+1)^2}\right| = -u + C_1$$

Exponentiate both sides:

$$\left|\frac{x}{(u+1)^2}\right| = e^{C_1} e^{-u} \implies \frac{(u+1)^2}{|x|} = C_2 e^u \implies (u+1)^2 = C x e^u$$

Step 7: Back-Substitute $u = \frac{y}{x}$

$$\left(\frac{y}{x} + 1\right)^2 = C x e^{y/x}$$

$$\left(\frac{x+y}{x}\right)^2 = C x e^{y/x} \implies \frac{(x+y)^2}{x^2} = C x e^{y/x}$$

Multiply both sides by x^2 :

$$(x+y)^2 = C x^3 e^{y/x}$$

Step 8: Check for Singular / Lost Solutions

During separation, we divided by $(u+1) = 0 \implies u = -1 \implies y = -x$. Check $y = -x$ in original ODE:

$$(x^2 + (-x)^2)dx + (x^2 - x(-x))(-dx) = (2x^2)dx + (2x^2)(-dx) = 0 \equiv 0$$

$y = -x$ is indeed a valid solution! In the final implicit relation, setting $(x + (-x))^2 = 0$ requires $c = 0$. Thus, $y = -x$ is included when $c = 0$.

4. Method 2: Bernoulli's Differential Equation

Standard Canonical Form

A first-order non-linear differential equation of the form:

$$\frac{dy}{dx} + P(x)y = f(x)y^n$$

where $n \in \mathbb{R}$, is called **Bernoulli's Equation**.

Inspection of Cases:

- If $n = 0$: $\frac{dy}{dx} + P(x)y = f(x) \implies$ **Linear 1st-Order ODE** (solved via $\mu(x) = e^{\int P dx}$).
 - If $n = 1$: $\frac{dy}{dx} + [P(x) - f(x)]y = 0 \implies$ **Separable 1st-Order ODE**.
 - If $n \neq 0$ and $n \neq 1$: The equation is **strictly non-linear**.
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The Power-Shift Substitution: $u = y^{1-n}$

The Derivation of the Linearization:

1. Divide the entire Bernoulli equation by y^n :

$$y^{-n} \frac{dy}{dx} + P(x)y^{1-n} = f(x)$$

2. Define the new dependent variable:

$$u = y^{1-n}$$

3. Differentiate u with respect to x using the chain rule:

$$\frac{du}{dx} = (1-n)y^{-n} \frac{dy}{dx} \implies y^{-n} \frac{dy}{dx} = \frac{1}{1-n} \frac{du}{dx}$$

4. Substitute into the equation:

$$\frac{1}{1-n} \frac{du}{dx} + P(x)u = f(x)$$

5. Multiply through by $(1 - n)$:

$$\frac{du}{dx} + (1 - n)P(x)u = (1 - n)f(x)$$

[!TIP] **Exam Golden Formula:** Memorize this single line! It transforms *any* Bernoulli equation into a standard first-order linear ODE in one step without repeating the intermediate chain-rule algebra.

Example 3: Complete Derivation (Lecture 5 Slide 11)

Problem Statement

Solve the differential equation:

$$x \frac{dy}{dx} + y = x^2 y^2$$

Step-by-Step Solution

Step 1: Put in Canonical Bernoulli Form

Divide through by the leading coefficient x (for $x \neq 0$):

$$\frac{dy}{dx} + \frac{1}{x}y = xy^2$$

Identify the key components:

$$P(x) = \frac{1}{x}, \quad f(x) = x, \quad n = 2$$

Since $n = 2 \neq 0, 1$, this is a non-linear Bernoulli equation.

Step 2: Define the Substitution

$$u = y^{1-n} = y^{1-2} = y^{-1} = \frac{1}{y}$$

Step 3: Transform into a Linear ODE

Using $(1 - n) = 1 - 2 = -1$:

$$\frac{du}{dx} + (-1)P(x)u = (-1)f(x)$$

$$\frac{du}{dx} - \frac{1}{x}u = -x$$

This is a standard first-order linear ODE in the variable $u(x)$!

Step 4: Compute the Integrating Factor $\mu(x)$

$$P_u(x) = -\frac{1}{x}$$

$$\mu(x) = e^{\int -\frac{1}{x} dx} = e^{-\ln|x|} = e^{\ln(|x|^{-1})} = \frac{1}{|x|} \implies \mu(\mathbf{x}) = \frac{1}{\mathbf{x}} \quad (\text{for } x > 0)$$

Step 5: Multiply and Integrate

Multiply the linear ODE by $\frac{1}{x}$:

$$\frac{1}{x} \frac{du}{dx} - \frac{1}{x^2} u = -1$$

Recognize the product rule on the LHS:

$$\frac{d}{dx} \left[\frac{1}{x} u \right] = -1$$

Integrate both sides with respect to x :

$$\frac{1}{x} u = \int -1 dx = -x + C$$

Multiply by x to isolate $u(x)$:

$$u(x) = x(-x + C) = Cx - x^2$$

Step 6: Back-Substitute $u = \frac{1}{y}$

$$\frac{1}{y} = Cx - x^2$$

Take the reciprocal to find $y(x)$:

$$y(x) = \frac{1}{Cx - x^2} = \frac{1}{x(C - x)}$$

Step 7: The Trivial Solution

Notice that $y(x) \equiv 0$ satisfies the original equation: $x(0) + 0 = x^2(0)^2 \implies 0 = 0$. Because $y = 0$ would cause division by zero in $u = 1/y$, it cannot be obtained from the general family for any finite constant C .

$$\text{Complete Solution: } y(x) = \frac{1}{Cx - x^2} \quad \text{and the singular solution } y(x) \equiv 0$$

5. Method 3: Reduction to Separation of Variables

The Linear Argument Form

Differential equations where the derivative depends strictly on a linear combination of x and y :

$$\frac{dy}{dx} = f(Ax + By + C) \quad (B \neq 0)$$

can **always** be transformed into an elementary separable equation.

The Substitution Rule

1. Define the substitute variable as the entire linear argument:

$$u = Ax + By + C$$

2. Differentiate implicitly with respect to x :

$$\frac{du}{dx} = A + B \frac{dy}{dx} \implies \frac{dy}{dx} = \frac{1}{B} \left(\frac{du}{dx} - A \right)$$

3. Substitute into the original equation:

$$\frac{1}{B} \left(\frac{du}{dx} - A \right) = f(u)$$

4. Isolate $\frac{du}{dx}$:

$$\frac{du}{dx} - A = Bf(u) \implies \frac{du}{dx} = A + Bf(u)$$

5. The resulting equation is completely **separable**:

$$\frac{du}{A + Bf(u)} = dx$$

Example 4: Complete Derivation (Lecture 5 Slide 12)

Problem Statement

Solve the differential equation:

$$\frac{dy}{dx} = (-2x + y)^2 - 7$$

Step-by-Step Solution

Step 1: Identify the Linear Combination

The right-hand side is a function of $(-2x + y)$, which matches $Ax + By + C$ with:

$$A = -2, \quad B = 1, \quad C = 0$$

Step 2: Define the Substitution

Let:

$$u = -2x + y$$

Differentiate with respect to x :

$$\frac{du}{dx} = -2 + \frac{dy}{dx} \implies \frac{dy}{dx} = \frac{du}{dx} + 2$$

Step 3: Substitute and Separate Variables

Substitute $\frac{dy}{dx} = \frac{du}{dx} + 2$ and $(-2x + y) = u$ into the ODE:

$$\frac{du}{dx} + 2 = u^2 - 7$$

Subtract 2 from both sides:

$$\frac{du}{dx} = u^2 - 9$$

This is a clean autonomous separable ODE!

Step 4: Constant Solutions Pre-Check

Before dividing by $u^2 - 9$, find where it equals zero:

$$u^2 - 9 = 0 \implies (u - 3)(u + 3) = 0 \implies u = 3 \quad \text{and} \quad u = -3$$

Since $u = -2x + y$:

- $u = 3 \implies -2x + y = 3 \implies y = 2x + 3$
- $u = -3 \implies -2x + y = -3 \implies y = 2x - 3$ Both are valid straight-line solutions to the ODE!

Step 5: Separate Variables & Partial Fractions

For $u \neq \pm 3$:

$$\frac{du}{u^2 - 9} = dx \implies \frac{du}{(u - 3)(u + 3)} = dx$$

Using partial fraction decomposition:

$$\frac{1}{(u - 3)(u + 3)} = \frac{A}{u - 3} + \frac{B}{u + 3} \implies 1 = A(u + 3) + B(u - 3)$$

- For $u = 3$: $1 = 6A \implies A = \frac{1}{6}$
- For $u = -3$: $1 = -6B \implies B = -\frac{1}{6}$

$$\frac{1}{6} \left(\frac{1}{u - 3} - \frac{1}{u + 3} \right) du = dx$$

Step 6: Integrate Both Sides

$$\frac{1}{6} \int \left(\frac{1}{u-3} - \frac{1}{u+3} \right) du = \int dx$$

$$\frac{1}{6} (\ln |u-3| - \ln |u+3|) = x + C_1$$

Multiply by 6 and combine logarithms:

$$\ln \left| \frac{u-3}{u+3} \right| = 6x + 6C_1$$

Exponentiate:

$$\left| \frac{u-3}{u+3} \right| = e^{6C_1} e^{6x} \implies \frac{u-3}{u+3} = ce^{6x} \quad (c = \pm e^{6C_1} \neq 0)$$

Step 7: Solve Explicitly for $u(x)$

$$u-3 = ce^{6x}(u+3) = ce^{6x}u + 3ce^{6x}$$

$$u - ce^{6x}u = 3 + 3ce^{6x}$$

$$u(1 - ce^{6x}) = 3(1 + ce^{6x})$$

$$u(x) = \frac{3(1 + ce^{6x})}{1 - ce^{6x}}$$

Step 8: Back-Substitute $u = -2x + y$

$$-2x + y = \frac{3(1 + ce^{6x})}{1 - ce^{6x}}$$

$$y(x) = 2x + \frac{3(1 + ce^{6x})}{1 - ce^{6x}}$$

Step 9: Verify Equilibrium Inclusion

- Setting $c = 0 \implies y = 2x + \frac{3(1)}{1} = 2x + 3$ (Included in the family!).
- Setting $u = -3 \implies \frac{3(1+ce^{6x})}{1-ce^{6x}} = -3 \implies 1 + ce^{6x} = -1 + ce^{6x} \implies 1 = -1$ (Impossible for finite c).
- Therefore, $y = 2x - 3$ is a **singular (lost) solution**.

6. Comprehensive Comparative Matrix & Exam Traps

Rapid Method Comparison Table

Feature	Homogeneous Equations	Bernoulli's Equations	Reduction to Separable
Standard Form	$M(x, y)dx + N(x, y)dy = 0$ with M, N deg α	$\frac{dy}{dx} + P(x)y = f(x)y^n$	$\frac{dy}{dx} = f(Ax + By + C)$
Substitution	$y = ux \implies dy = udx + xdu$	$u = y^{1-n}$	$u = Ax + By + C$
Derivative Relation	$\frac{dy}{dx} = u + x \frac{du}{dx}$	$\frac{du}{dx} = (1-n)y^{-n} \frac{dy}{dx}$	$\frac{du}{dx} = A + B \frac{dy}{dx}$
Transformed Form	Separable in u and x	Linear in $u(x)$	Separable in u and x
Key Algebraic Step	Factor out x^α and cancel	Divide standard form by y^n	Isolate $\frac{du}{dx} = A + Bf(u)$

Top 5 Concordia Traps for Lecture 5 on Quizzes and Midterms

- Forgetting to Standardize Bernoulli Before Identifying $P(x)$:**
 - Given $xy' + y = x^2y^2$, do NOT say $P(x) = 1$! You MUST divide by x first: $y' + \frac{1}{x}y = xy^2 \implies P(x) = \frac{1}{x}$.
- The Product Rule in $y = ux$:**
 - Never write $dy = u dx$. You must apply the product rule: $dy = u dx + x du$.
- The Constant C in Bernoulli General Solutions:**
 - When solving the linear equation for u , remember: $u(x) = \frac{1}{\mu(x)} \int \mu Q dx + \frac{C}{\mu(x)}$. You must add C before dividing by $\mu(x)$, and back-substitute $y = u^{1/(1-n)}$ to the entire expression!
- Neglecting Trivial and Singular Solutions:**
 - For Bernoulli with $n > 0$, $y(x) \equiv 0$ is always a solution that is divided out during substitution.
 - For linear arguments, $u^2 - a^2 = 0 \implies u = -a$ produces a straight-line singular solution (e.g. $y = 2x - 3$).
- Sign Errors in $\frac{du}{dx} = A + B \frac{dy}{dx}$:**
 - If $u = 2x - y + 1$, then $\frac{du}{dx} = 2 - \frac{dy}{dx} \implies \frac{dy}{dx} = 2 - \frac{du}{dx}$. Missing the minus sign reverses the entire integral!